

Hybrid Van der Pol–Duffing Oscillator Workbench

User Manual & Technical Reference

Version 1.0

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Quick Start

```
% In MATLAB Command Window:  
>> Hybrid_VDP_Duffing_GUI
```

Select parameters → click **Build** → click **Simulate** → inspect plots.

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1 Introduction

The **Hybrid VDP-Duffing Oscillator Workbench** is a self-contained MATLAB/Simulink/Simscape application that simulates and analyses the dynamics of the parametrically-forced Hybrid Van der Pol–Duffing (HVDPD) nonlinear oscillator. The oscillator equations are solved numerically inside Simulink, while two Simscape RC sub-networks emulate fractional-order impedance elements (Foster-I representation).

1.1 Key capabilities

- Interactive parameter panel with real-time computed parameter readout.
- Three build modes: *Simulink Only*, *Deploy to Arduino Due*, and *USB Scope Control Only*.
- Live phase portrait, state-vs-forcing time-series, and Hann-windowed power spectrum.
- Advanced plots: Poincaré sections, spectrograms, and peak-to-peak return maps.
- High-resolution export (PNG / JPEG / MATLAB FIG) for all plots and Simulink model diagrams.
- Arduino Due hardware-in-the-loop (HIL) DAC streaming at up to 10 kHz.

1.2 System requirements

Component	Minimum Version
MATLAB	R2022a (9.12)
Simulink	10.5
Simscape	5.3
Simscape Electrical	7.7
<i>Optional (HIL)</i>	
Simulink Support Package for Arduino Hardware	any
Embedded Coder	any

2 Mathematical Background

2.1 The Hybrid VDP-Duffing equation

The oscillator is governed by the second-order ODE:

$$\ddot{x} - d(1 - x^2)\dot{x} + \omega_0^2 x + \beta x^3 = f \cos(\Omega t) [1 + h \cos(\omega_p t)] \quad (1)$$

where:

Symbol	Name	Unit
d	Damping coefficient (Van der Pol)	—
ω_0	Natural frequency	rad/s
β	Duffing nonlinearity coefficient	—
f	Forcing amplitude	—
Ω	Driving frequency	rad/s
h	Modulation depth	—
ω_p	Parametric modulation frequency	rad/s

Eq. (1) is written as the first-order state system:

$$\dot{x} = y \quad (2)$$

$$\dot{y} = d(1 - x^2)y - \omega_0^2 x - \beta x^3 + f \cos(\Omega t) [1 + h \cos(\omega_p t)] \quad (3)$$

2.2 Resistor-ratio parameterisation

The circuit constants are derived from a single base resistance R and capacitance C :

$$d = \frac{R}{R_1}, \quad \omega_0^2 = \frac{R}{R_2}, \quad \beta = \frac{R}{R_3} \quad (4)$$

This maps every coefficient to a physical resistor ratio, facilitating direct hardware realisation.

2.3 Simscape RC fractional-order emulation

Each RC sub-network (RC_X for the x -channel, RC_Y for the y -channel) implements a parallel RC branch:

Block	Role
Controlled Current Source	Receives commanded current $I_{\text{cmd}} = C\dot{x}$ or $C\ddot{x}$
Capacitor C	Stores fractional state; outputs voltage V_C
Resistor R_{leak}	Parallel leakage (sets time-decay constant $\tau = RC$)
Voltage Sensor	Feeds V_C back as Simulink signal
Solver Configuration	Provides Simscape network integrator
Electrical Reference	Ground node

The Foster-I equivalent circuit of a fractional capacitor of order α is approximated by N parallel RC branches. In the current 2-branch implementation ($N = 2$), one branch per state variable is used.

3 Installation

3.1 From the .mltbx file (recommended)

```
% Option A -- double-click Hybrid_VDP_Duffing_Toolbox_v1.0.mltbx
%               in Windows Explorer
% Option B -- from MATLAB Command Window:
```

```
matlab.addons.install('Hybrid_VDP_Duffing_Toolbox_v1.0.mltbx')
```

MATLAB will add the toolbox to the search path automatically.

3.2 Manual installation (source folder)

```
addpath('d:\path\to\Hybrid_VDP_Duffing_GUI_2RC_v1');
savepath;
```

3.3 Building the .mltbx yourself

```
cd('d:\path\to\Hybrid_VDP_Duffing_GUI_2RC_v1')
package_toolbox()
% Produces: Hybrid_VDP_Duffing_Toolbox_v1.0.mltbx
```

4 GUI Overview

Launch the workbench with:

```
Hybrid_VDP_Duffing_GUI
```

The window is divided into three regions:

Region	Content
Left panel	All simulation parameters, circuit constants, excitation, hardware settings, and run setup.
Right panel	Three live plot axes (phase portrait, state vs forcing, power spectrum) and action buttons.
Top bar	Developer credits.

4.1 Action buttons

Button	Action
Clean	Closes all Simulink models and clears workspace variables from previous runs.
Debug	Prints the current parameter values to the MATLAB Command Window for inspection.
Build	Generates the Simulink/Simscape model according to the selected build mode.
Simulate	Runs the model and updates all plots.
Stop	Stops a running simulation gracefully.
Export...	Opens the Results Export dialog.
Model Shot...	Opens the Simulink Screenshot Export dialog.

Typical workflow: Build → Simulate → Export.

5 Parameter Reference

5.1 Mode

Build Mode	Description
Simulink Only	Variable-step <code>ode15s</code> solver. No hardware required. Fastest iteration.
Deploy to Arduino Due	Fixed-step <code>ode4</code> (RK4) code generation. Streams DAC output to the Due.
USB Scope Control Only	Simulink-only model but with oscilloscope control blocks.

5.2 Circuit Constants

Field	Default	Meaning
Base R (Ω)	10 k Ω	Fixed reference resistance; all coefficients are ratios to this value.
Capacitance C (F)	100 nF	Simscape capacitor value. Scales the RC time constant.
R1 (for $d = R/R_1$)	800 k Ω	Sets Van der Pol damping $d = 0.0125$.
R2 (for $\omega_0^2 = R/R_2$)	10 k Ω	Sets natural frequency $\omega_0 = 1$ rad/s.
R3 (for $\beta = R/R_3$)	30 k Ω	Sets Duffing coefficient $\beta \approx 0.333$.

Click **Update Computed Parameters** to refresh the readout labels d , ω_0 , β , and f/Ω^2 after changing any circuit constant.

5.3 Excitation

Field	Default	Meaning
Forcing Amplitude (f)	140	Amplitude of the cosine driving term.
Modulation Depth (h)	0.2	Depth of the parametric envelope (slider: 0.1–1).
Parametric Freq (ω_p)	3 rad/s	Frequency of the modulation envelope (slider: 0.1–5).
Driving Freq (Ω)	10 rad/s	Frequency of the cosine carrier (slider: 1–10).

5.4 Hardware Output (Simscape)

Field	Default	Meaning
Leak Resistor Rx (Ω)	1 G Ω	Parallel leakage on the RC_X network. Very large = near-ideal capacitor.
Leak Resistor Ry (Ω)	1 G Ω	Parallel leakage on the RC_Y network.

5.5 Arduino Due DAC Output

These controls are active only in *Deploy to Arduino Due* mode.

Control	Description
Enable DAC Output	Adds the DAC output branch to the model.
Enable XY DAC1 = $y(t)$	Routes $y(t)$ to the second DAC channel for XY scope display.
Smooth DAC Output	Adds a first-order low-pass filter before the DAC.
Physical View	Selects the oscilloscope display mode.
Sample Time (s)	DAC update rate (min 100 μ s).
LPF tau (s)	Time constant for the optional smoothing filter.
Amp x	Physical amplitude multiplier before DAC scaling.
Min / Max	Logical signal range mapped to 0–3.3 V on the Due DAC.

5.6 Run Setup

Field	Default	Meaning
Simulation Time (s)	600	Total integration time.
Init Pos (x_0)	−1	Initial displacement.
Init Vel ($\dot{x}_0 = y_0$)	1	Initial velocity.

6 Plot Panel

After a successful simulation, three plots update automatically.

6.1 Phase Portrait

Shows the trajectory $(x(t), y(t))$ in state space.

- **Slate-blue** — transient portion (first 35% of simulation).
- **Crimson** — steady-state attractor (remaining 65%).
- **Amber circle** — initial condition (x_0, y_0) with an annotated arrow.
- **Red arrow** — points to a representative attractor point.

6.2 State vs Parametric Forcing

Time-domain overlay of $x(t)$ (navy) and $h \cos(\omega_p t)$ (orange dashed). A shaded region marks the transient window; an arrow marks the onset of steady state.

6.3 Power Spectrum

Hann-windowed single-sided amplitude spectrum of $x(t)$ in dB. A dashed vertical line indicates the driving frequency Ω ; an annotated arrow points to the dominant spectral peak.

7 Advanced Plots

These standalone figure windows are accessed from the **Export** dialog or via the following buttons (if added to the toolbar):

7.1 Poincaré Section

Samples $(x(nT), y(nT))$ at integer multiples of the driving period $T = 2\pi/\Omega$. A scattered cloud indicates quasi-periodic or chaotic motion; isolated points indicate period- n orbits.

7.2 Spectrogram

Short-time Fourier transform of $x(t)$. Reveals time-evolution of frequency content.

7.3 Return Map (Peak-to-Peak)

Successive peak amplitudes x_n vs. x_{n+1} . Period-doubling cascades appear as bifurcating branches.

8 Export Functionality

8.1 Results Export (Export... button)

Option	Output
Phase portrait	Single-axes PNG/JPEG/FIG.
State vs forcing	Single-axes PNG/JPEG/FIG.
Power spectrum	Single-axes PNG/JPEG/FIG.
Combined tiled	2×2 layout of all three plots.
Poincaré section	Separate figure window exported.
Spectrogram	Separate figure window exported.
Return map	Separate figure window exported.

Supported formats: **PNG** (default, 600 DPI), **JPEG**, **MATLAB FIG**.

8.2 Model Screenshot (Model Shot... button)

Captures the Simulink block diagram as a high-resolution image.

- **Top-level model** — shows ODE_Core subsystem and I/O blocks.
- **RC_X subsystem** — Simscape RC circuit for the x channel.
- **RC_Y subsystem** — Simscape RC circuit for the y channel.

Set zoom percentage and tick *Fit to page* for publication-ready captures.

9 Arduino Due HIL Mode

9.1 Setup

1. Install the *Simulink Support Package for Arduino Hardware* from MATLAB Add-On Explorer.

2. Connect the Arduino Due via its **programming USB port** (not the native port).
3. Set Build Mode to **Deploy to Arduino Due**.
4. Click **Build** — the model is generated and configured.
5. Click **Simulate** (or use the Simulink *Deploy* action) to flash.

9.2 COM-port override

If auto-detection fails, force the port manually before clicking Build:

```
% In MATLAB Command Window:
arduino_due_com_port_override = 'COM7'; % replace with your port
```

9.3 DAC scaling

The Due DAC outputs 0–3.3 V (12-bit, 4096 levels). The GUI maps the logical range [Min, Max] linearly to this physical range:

$$\text{Code} = \text{round}\left(\frac{x_{\text{logical}} - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}} \cdot 4095\right) \quad (5)$$

For XY oscilloscope display, connect DAC0 (*x*-channel) and DAC1 (*y*-channel) to the oscilloscope inputs.

10 Troubleshooting

Symptom	Remedy
<i>Solver Configuration not connected</i> error during Simulate	Click Clean then Build again. Each RC subsystem must contain its own Solver and GND blocks.
Export fails with <i>Invalid parameter/value pair</i>	Update to toolbox v1.0 – this was a known bug in the <code>copy_axes_contents</code> function (fixed).
<i>Arduino Due was not detected</i> dialog on Build	Set <code>arduino_due_com_port_override</code> in MATLAB workspace. Build still completes with a placeholder port.
Phase portrait shows only noise	Increase Simulation Time or reduce f . The transient window is 35% of total time; very short runs may not reach steady state.
Power spectrum shows no peaks	Ensure $\Omega \leq 10$ rad/s (default x-axis limit). Use <code>xlim</code> on the spectrum axes if $\Omega > 10$.
<i>SizeChangedFcn</i> warning on startup	Fixed in v1.0 (<code>AutoResizeChildren</code> set to <code>off</code>). Ignore if using an older copy.

11 Mathematical Appendix

11.1 Steady-state window

The code defines the steady-state index set as:

$$\mathcal{S} = \{i : t_i \geq t_1 + 0.35(t_N - t_1)\}$$

i.e. the last 65% of the simulation is treated as steady state for spectral analysis and attractor rendering.

11.2 Hann-windowed FFT

The amplitude spectrum uses a Hann (raised-cosine) window to reduce spectral leakage:

$$w[n] = \frac{1}{2} \left(1 - \cos \frac{2\pi n}{L-1} \right), \quad n = 0, \dots, L-1$$

The single-sided magnitude (in dB) is:

$$|X(\omega_k)|_{\text{dB}} = 20 \log_{10} \left(\frac{2|X_k|}{L} + \epsilon \right)$$

where $\epsilon = 10^{-12}$ prevents $\log(0)$.

11.3 Poincaré section timing

Samples are taken at times $t_n = t_{\text{start}} + nT$, where $T = 2\pi/\Omega$ and t_{start} is chosen to begin after the first 35% of the run. Inter-sample values are obtained by `interp1` (linear). At least 8 samples are required.

11 Acknowledgements

This toolbox was developed as part of ongoing doctoral research into fractional-order nonlinear dynamical systems. Hardware validation was performed using GW Instek oscilloscopes and Arduino Due prototyping boards.

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